

Assignment 6

Program -

```
1]
function queue()
{
    this.data=[];
    this.rear=0;
    this.front=0;
    this.enqueue=enqueue;
    this.MaxSize=10;
    this.print=print;
    this.length=length;
    this.isEmpty=isEmpty;
    this.dequeue=dequeue;
}
function enqueue(element)
{
    if(this.rear<this.MaxSize)
    {
        this.data[this.rear++]=element; //data[0]=10
    }
}

function print()
{
    for(let i=0;i<this.data.length;i++)
    {
        console.log(this.data[i])
    }
}
function length()
{
    return this.rear;
}
function isEmpty()
{
    return this.rear=0;
}
function dequeue()
{
    if(this.isEmpty()==false)
    {
        this.front=this.front+1;
        return this.data.shift();
    }
}

var q=new queue();
q.enqueue(10);
q.enqueue(20);
q.enqueue(30);
console.log("elements in queue :");
q.print();
```

```
console.log("length of queue :",q.length());  
console.log("delete one element of queue is :")  
console.log(q.dequeue())  
console.log("elements in queue :");  
q.print();
```

Output -

```
elements in queue :  
10  
20  
30  
length of queue : 3  
delete one element of queue is :  
10  
elements in queue :  
20  
30
```